

Nano-Optics for Depth Sensing

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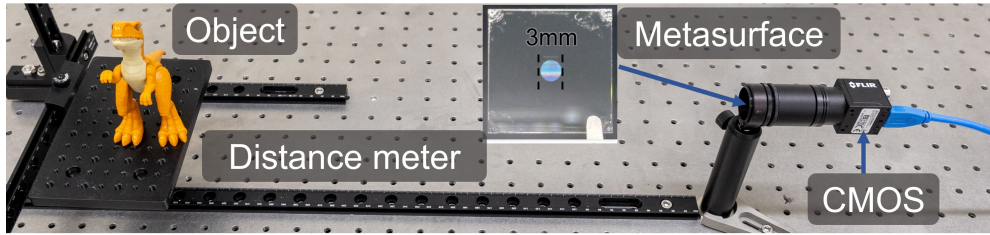
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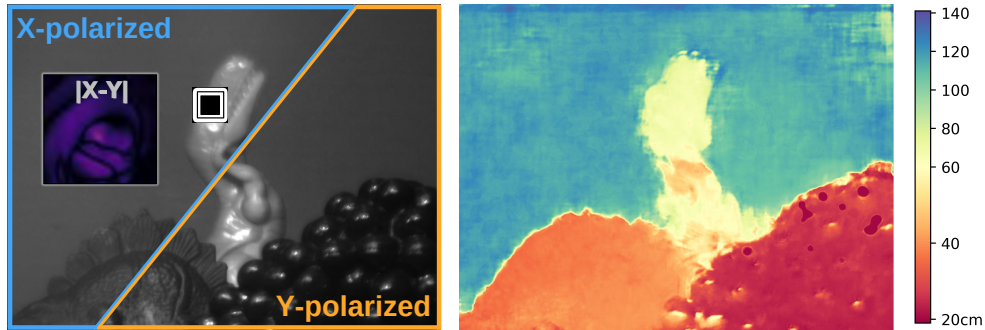
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(a) demo setup of depth imaging system



(b) metasurface-polarized image pair

(c) depth prediction

Figure 1: Demo setup for interactive depth capture. (a) shows our demo setup. The metasurface, along with a 590-nm bandpass filter, is mounted in a 1-inch lens tube with a monochrome CMOS sensor to capture the tabletop. The inset shows our 3-mm diameter metasurface with a thickness of only 700 nm on a 0.5-mm-thick glass substrate, which will be inspected by participants under a microscope. (b) A metasurface-polarized image pair with the zoomed-in inset visualizing their local pixel-wise disparity. (c) A representative depth imaging result, obtained through a single-shot capture with our Nano-3D system. Participants are encouraged to rearrange a set of objects and observe the real-time depth map. The unit of the color bar is in centimeters.

Abstract

Depth imaging powers applications from robotics to immersive VR/AR, yet conventional solutions often rely on bulky hardware or inaccurate approximations. We present Nano-3D, an ultra-compact metasurface-based neural depth sensor that captures single-shot, orthogonally polarized image pairs and reconstructs metric depth in real time. At SIGGRAPH 2025, participants can place 3D-printed objects of various shapes and sizes before Nano-3D to watch live update of robust depth maps, even under challenging conditions. A parallel display comparing a state-of-the-art monocular depth estimation model will be provided to highlight Nano-3D's superior accuracy. Participants can also inspect the 700-nm-thick TiO₂ metasurface chip under a microscope, discovering the nanophotonic structures that powers Nano-3D.

CCS Concepts

• Hardware → Metasurface.

Keywords

Metasurface, Depth Imaging

ACM Reference Format:

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1 Introduction

Accurately capturing metric depth is fundamental for numerous applications in graphics, robotics, and immersive technologies [Lindell et al. 2018; Rogers et al. 2021]. Conventional active systems (e.g., time-of-flight LiDAR) and multi-lens stereo solutions often suffer from bulky footprints or complex architecture. Recently, metasurfaces have emerged as an ultra-compact solution, offering precise subwavelength control of light [Kuznetsov et al. 2024; Yu and Capasso 2014]. In Nano-3D, we co-design a custom-fabricated TiO₂ metasurface and a multi-module deep neural network to achieve physically accurate, single-shot, monocular metric depth sensing from orthogonally polarized image pairs. We demonstrate how our approach captures high-quality depth map, using 3D-printed objects of different sizes and shapes placed at varying distances, which outperforms purely software-based monocular depth estimation.

2 Nano-3D Architecture and Implementation

Metasurface Design and Fabrication. We fabricate a 700-nm-thick TiO₂ metasurface consisting of dense nanopillars arranged in a sub-wavelength square lattice. By leveraging birefringent meta-atoms with engineered cross-sections, we independently control the phase

for the x - and y -polarization channels. This creates two conjugate rotating point spread functions (PSFs), each shifting in opposite directions as a function of object depth. Our metasurface has a diameter of 3 mm and operates at a center wavelength of 590 nm.

Dual-Polarization Depth Encoding. As shown in Figure 2, when unpolarized light from a 3D scene enters the metasurface, each 3D point is split into two polarized images on the CMOS sensor. Depth information is encoded via the relative rotation between the image pair.

Neural Depth Decoding. A lightweight neural network first calculates a PSF shift map by locally matching features in the polarized image pair. Then, a U-Net-style depth decoder decodes pixel-wise metric depth from this shift map, with additional context from the polarized image pair.

3 Results

We evaluated Nano-3D both in simulation and on 28 physical scenes containing objects of varying sizes and distances. On a novel synthetic dataset with randomly placed geometry, Nano-3D outperformed leading monocular depth estimators with a model size 80% smaller. In physical tests, it handled real-world occlusions more robustly than an Intel RealSense D455, thanks to its monocular nano-optics design and minimal image pair disparity. We also verified that measured point spread functions closely matched simulator predictions, confirming the accuracy of our hardware-software co-design and highlighting the viability of nano-optics depth sensing in complex scenarios.

4 User Experience and Demonstration

Participants will be invited to interact with a table of 3D-printed objects of various shapes and scales, such as architectural miniatures or creature figurines, as shown in Figure 1a. A compact Nano-3D camera, mounted to overlook this tabletop scene, continuously captures single-shot, orthogonally polarized image pairs of the rear-ranged objects. Participants will see a live depth map generated by our Nano-3D system, which maintains metric accuracy even when users place identical models of different sizes at varying distances. Real-time results from a state-of-the-art monocular depth estimation model will be displayed in parallel for comparison, highlighting the difficulty of preserving accurate depth in such tricky scenarios.

A microscope station will showcase the actual metasurface chip, allowing participants to observe the specially engineered TiO₂ structures that power Nano-3D's unique polarization depth encoding. By freely reconfiguring the 3D-printed objects and comparing the dual-screen outputs in real-time, participants gain firsthand insight into how Nano-3D's hardware-software co-design enables precise depth sensing with a 700 nm thick optical component, offering a glimpse into future compact, high-fidelity 3D sensors.

5 Conclusion

We introduce Nano-3D, an ultra-compact metasurface-based neural depth imager that achieves physically accurate, single-shot depth sensing. Our fabricated TiO₂ metasurface works in tandem with a specialized neural network to decode precise pixel-wise depth from orthogonally polarized image pairs. We demonstrate Nano-3D's

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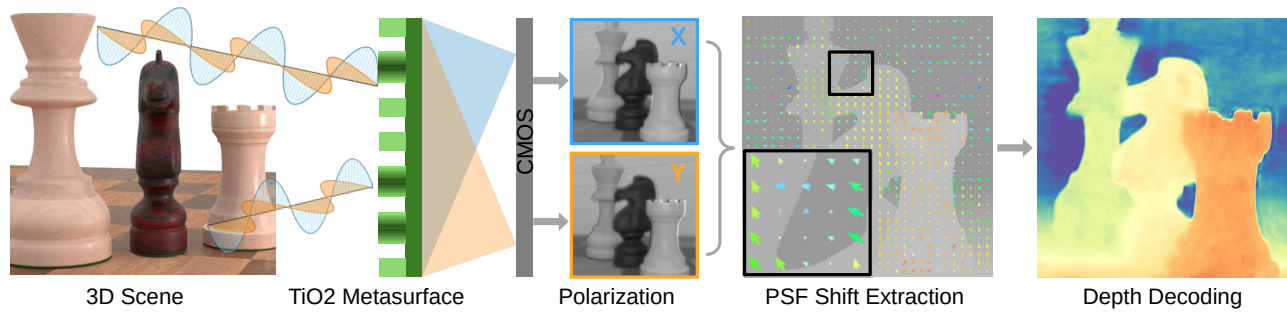


Figure 2: Overview of our Nano-3D sensing-computation framework. In the point spread function (PSF) shift extraction, the pixel-wise shifts are visualized by colored arrows, where the color indicates shift direction and arrow length indicates shift magnitude. In the depth decoding, warmer colors indicate closer distances.

effectiveness on both simulated scenes and real tabletop objects of varying size and complexity. We believe this fusion of nano-optics and computational imaging opens exciting new avenues for next-generation AR/VR, mobile robotics, and advanced display systems.

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